B.TECH CSE (ML / AI) · MAJOR PROJECT · WEEK 2 OF 20

MFSTNet — Camera-Only Congestion Forecasting and Signal Control

A multimodal deep-learning system that forecasts per-lane congestion 60 seconds ahead from traffic camera video alone, and a reinforcement-learning controller that acts on it — built for unstructured Indian traffic, where lane discipline cannot be assumed.

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0.8941
mAP50, elevated CCTV

629
committed tests

18
decisions recorded

1.5 h
full ablation

Every validation figure is read from a committed results CSV at build time. The two exceptions — S17's detector numbers and the timing comparison — say so where they appear.

Lane discipline is an assumption this traffic does not make

Most traffic forecasting assumes vehicles in lanes, of similar size, detected by sensors in the road. Every one of those assumptions fails here.

- **Vehicles do not stay in lanes.**
  Motorcycles filter between cars, auto-rickshaws occupy gaps. A lane occupancy figure from a loop detector describes a road this is not.
- **The vehicle mix is extreme.**
  A motorcycle and a bus differ by an order of magnitude in road space. Counting vehicles treats them as equal; the measured fleet is 45% two-wheelers.
- **The infrastructure is not there.**
  Inductive loops need the road dug up and maintained. Cameras are already mounted at many junctions, and a phone on a footbridge is a valid sensor.
- **Off-the-shelf detection does not transfer.**
  COCO has no auto-rickshaw class at all, and a detector trained on dashcam imagery scores 0.3223 on an elevated camera.

WHAT THE SYSTEM HAS TO DO

- **Forecast, not report**
  `60 s ahead, per lane, three classes`
- **From pixels only**
  `no loops, no GPS, no V2I radio`
- **Survive an unseen camera**
  `the number that governs deployment`
- **Act on the forecast**
  `PPO controller consumes it directly`
- **Degrade safely**
  `counts alone, then a local Webster fallback`

A forecast nothing acts on is a dashboard. The controller is what turns a prediction into changed signal timing, which is why both halves are in scope.

Four subsystems, integrated late by design

Each is independently testable, which is why three of them are already measured while the fourth is still being decided.

DATASET

IndiaTrafficNet

8 India-specific classes
including auto- and
e-rickshaw, cattle

›

DETECTION

YOLOv8s

per-lane vehicle counts
0.8941 mAP50 at the
deployment viewpoint

›

THE MODEL

MFSTNet

ResNet-50 + DINOv2 ViT
gated cross-attention
per-lane forecast

›

CONTROL

PPO agent

SUMO via TraCI
consumes the forecast
as auxiliary state

THE COUPLING THAT IS EASY TO MISS

MFSTNet's output is not only a prediction — it occupies indices 11–14 of the PPO controller's 16-dimensional state vector. Changing the model's output shape invalidates every trained controller checkpoint.

Two fallbacks are required behaviour rather than nice-to-haves: if the model fails, the controller runs on raw counts with those indices zeroed — never a shortened vector; if the network drops for more than 10 seconds, the edge node switches to a Webster controller embedded locally.

Every version here is the one actually installed

Until four days ago these were aspirational: not one of the seven pinned-and-installed packages matched its pin.

VISION AND LEARNING

- torch 2.13.0+cu126
- torchvision 0.28.0
- timm 1.0.28 (DINOv2)
- ultralytics 8.4.118
- opencv 5.0.0.93

REINFORCEMENT LEARNING

- stable-baselines3 2.9.0
- gymnasium 1.3.0
- eclipse-sumo 1.27.1
- traci / sumolib 1.27.1

ANALYSIS AND TESTING

- numpy 2.5.2
- pandas 3.0.5
- scipy 1.18.0
- scikit-learn 1.9.0
- xgboost 3.4.1
- pytest 9.1.1

WHY SCIPY AND SCIKIT-LEARN ARE PINNED BUT NOT USED

The metrics and statistics are pure standard library so the arithmetic stays auditable. These two exist so the tests can cross-check the hand-rolled implementations against them — macro F1, quadratic-weighted kappa, the paired t-test and Cohen's d are graded numbers, and a bespoke statistic nobody verifies is worse than none.

Hardware: RTX 4050 Laptop, 6 GB, sm_89 · training runs locally, not on Colab

Seven sources, and what we may do with each

Licence is not paperwork here — it decides what can be published, and one of these is unresolved.

DATASET & SOURCE	LICENCE	ROLE IN THIS PROJECT
BMD-45 huggingface.co/datasets/iisc-aim/BMD-45	CC BY 4.0	Primary training and test at the deployment viewpoint
IDD Detection idd.insaan.iiit.ac.in — IIIT Hyderabad	Research use	Week-2 bootstrap; the project's only dashcam surface
TrafficCAM github.com/Math-ML-X/TrafficCAM	NONE DECLARED	Cross-camera test — and the only source of e_rickshaw
ITD v1.2 github.com/teg-iitr/ITD-Indian-traffic-dataset	CC BY-NC 4.0	Supplies the detector used for lane survey and counting
Bellevue Traffic Video github.com/City-of-Bellevue/TrafficVideoDataset	Public body release	ADR-016 Phase 1 corpus — proposed, not yet used
Public traffic footage 12 clips resolving to 11 distinct cameras	Third-party	Every MFSTNet training sequence comes from these
IndiaTrafficNet self-collected, self-annotated	Ours	ADR-006 — pending sign-off

BEFORE ANYTHING IS PUBLISHED

- **TrafficCAM: none declared**
  checked — repo, site, download page
- **IDD requires a citation**
  Varma et al., WACV 2019
- **ITD is non-commercial**
  CC BY-NC 4.0 propagates
- **BMD-45 is the clean one**
  CC BY 4.0, attribution only

No licence is not the same as permissive: with none declared, default copyright applies. Research use is normal practice; redistributing the data or releasing weights trained on it is not granted. The authors are named and reachable, so we ask them — and cite the paper either way.

Detection at the deployment viewpoint is solved

Milestone M2 was due in Week 9. It was met in Week 2, on BMD-45's Bengaluru CCTV test split.

CLASS	PRECISION	RECALL	MAP50	MAP50-95	TEST BOXES
car	0.8617	0.8989	**0.9212**	0.7849	2,770
motorcycle	0.865	0.8186	**0.9033**	0.674	5,910
auto_rickshaw	**0.8892**	**0.8654**	**0.9186**	**0.7635**	**2,132**
bus	0.8481	0.8105	**0.8832**	0.7155	606
truck	0.8032	0.7694	**0.8442**	0.6784	939
mean	**–**	**–**	**0.8941**	**–**	**12,357**

THE NUMBER THAT MATTERS

- **auto_rickshaw 0.9186**
  a class COCO does not have
- **e_rickshaw, cattle**
  absent from this test split

Building our own detector is justified by the classes an off-the-shelf model cannot see at all, not by a few points of mAP on the classes it can.

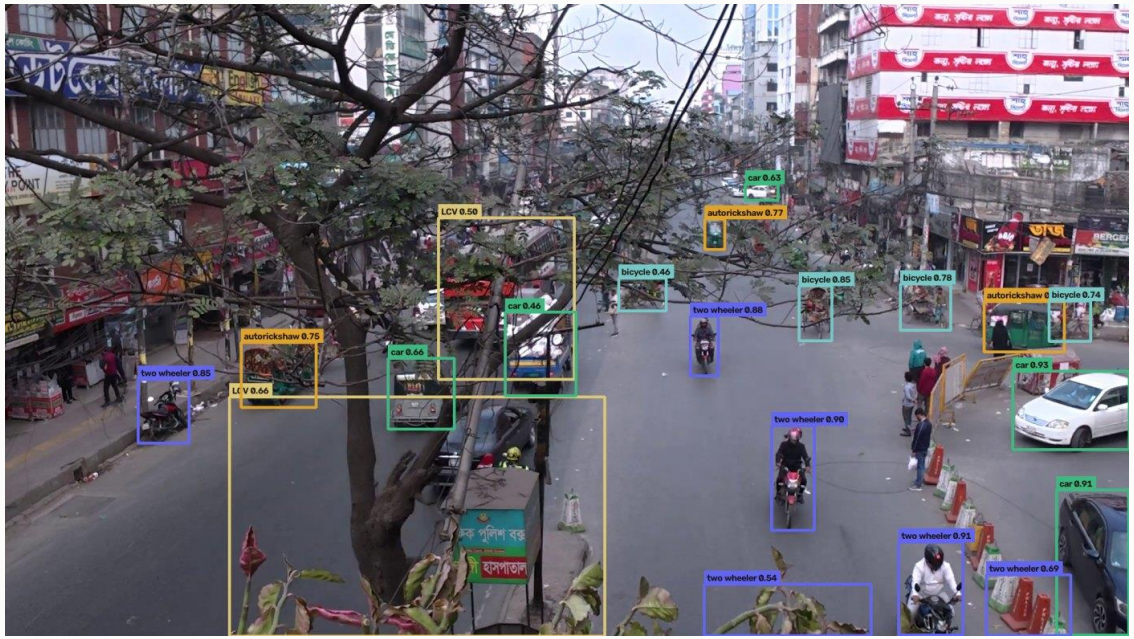
What "unstructured" actually means

Real output from the detector at confidence 0.45. Neither scene has lane markings a model could rely on.



Mumbai — Andheri Station

23 vehicles in one frame. A bus filling a quarter of the image and motorcycles under one percent of it.



Dhaka — Rampura

20 vehicles. The pilot camera: its quietest moment in twenty minutes still held nine vehicles.

Boxes coloured by class in the detector's own taxonomy · vehicles span 25% of the frame down to under 1%

+53.2% mAP over COCO on the shared classes

M2's criterion is a 10% improvement over COCO. It had been asserted and never measured, so we measured it — same test split, ground truth remapped so one validator scores both.

CLASS	COCO BASELINE	OURS	DELTA
car	0.6048	0.7133	+0.1085
motorcycle	0.4285	0.6784	+0.2499
bus	0.4852	0.7141	+0.2288
truck	0.3097	0.6958	+0.3861
shared mean	**0.4571**	**0.7004**	+53.2%

WHERE NO PERCENTAGE IS HONEST

- **auto_rickshaw 0.7109**
  COCO has no such class
- **cattle 0.3516**
  COCO has no such class

The baseline is not low here, it is undefined. Saying that is a stronger statement than any percentage would have been, and it is the difference between a result and a marketing number.

Counting accuracy is not detection accuracy

Congestion labels come from counts, and a detector with excellent mAP can still systematically over- or under-count. So the operating point was fitted against 5,273 human-annotated vehicles.

CONFIDENCE	DETECTED / TRUE	SPARSE THIRD	DENSE THIRD	MAE PER FRAME
0.25	1.1724	1.2147	1.1686	1.998
0.35	1.0793	1.1207	1.0754	1.349
0.45	**0.9989**	**1.0251**	**0.9968**	**1.072**
0.55	0.934	0.9608	0.9271	1.165
0.6	0.9006	0.9436	0.8917	1.361

WHY THE LAST TWO COLUMNS EXIST

- **0.45 is pinned**
  ratio 0.9989, lowest error
- **Density dependence 2.8%**
  reported, not averaged away

A detector accurate on average can still be biased with density — under-counting exactly when the road is busy, which is exactly when the forecast matters. Splitting the test set by density is the only way to see it.

The number that governs deployment is 0.3500, not 0.8941

A model deployed on a junction it has never seen is in the third column. Reporting only the first would be the easiest mistake in this project.

BMD-45 — TRAINED DOMAIN

0.8941

Bengaluru Safe City CCTV, the viewpoint the detector was built for

TRAFFICCAM — CAMERAS NEVER SEEN

0.3500

five other Indian cities, same elevated viewpoint, different cameras

the same weights lose 61% of their accuracy moving to cameras they were not trained on

THE COMPARISON THAT SHOWS IT

- **Dashcam-trained on CCTV**
  `0.3223`
- **Same model, own test set**
  `0.6201`

Viewpoint, not country, is what breaks a detector. The same weights lose half their accuracy moving from a dashcam to a pole, which is why the whole data strategy targets elevated cameras.

WHAT FIXED IT, AND WHY WE DID NOT ADOPT IT

Adding 1,254 human-annotated TrafficCAM boxes moved cross-camera accuracy to 0.5734 and brought e_rickshaw from 0.000 to 0.4787 — a class the detector had been shipping an output head for without ever seeing an example. That arm was recorded as failing a sixth criterion on IDD. Re-measuring the baseline it was judged against does not reproduce it, so the verdict is suspended rather than reversed: both sides have to be measured the same way before it means anything.

Nine steps, and the expensive one runs once

Turning video into training data. The detector pass costs hours; every choice made after it re-derives in seconds.

triage
which clips qualify

survey
where traffic is

place lanes
by hand, per camera

import
validate placement

build corpus
detector → counts

relabel
thresholds

audit
re-derive and check

cache
frozen features

train
config or ablation

WHAT A CHANGE ACTUALLY COSTS

Move a lane centre
0.66 s

Change the thresholds
seconds

Change detector, confidence or sampling
hours

A lane that moves when the tooling changes is not a property of the road

Lane centres are placed by a human over a detection heat map, and the reason is measurable.



MEASURED, NOT ASSUMED

- **0.164 of frame width**
  ≈ 315 px of drift
- **Between two detectors**
  same footage, same settings

Automatic centres are a clustering of where the detector happened to fire. Swap the detector and they move. A reviewer placing them by hand is placing something about the road instead.

The count matters more than the positions. At two centres, two cameras left over a third of vehicles outside every lane; at four, the worst is 10.5%.

Every 2 seconds, not 5, keeps four cameras alive

The PRD fixes a 5-second interval as a bare number derived nowhere. We pre-registered a statistic to replace it before the footage existed, deliberately biased against the answer we wanted.

STEP_S	CLIP MINIMUM	SEQUENCES	INDEPENDENT SEGMENTS	CAMERAS CONTRIBUTING
5 — current	355 s	302	29	7 of 11
2 — proposed	**178 s**	**359**	**64**	11 of 11
1	119 s	381	101	11 of 11

THE SEQUENCE COLUMN IS THE WEAKER ARGUMENT

Overlapping windows inflate any sequence count — measured at 5.6×. The column that matters is the last one: a 5-second interval needs 355 s of continuous footage per clip and so deletes four cameras outright. Sequences from one camera are correlated; cameras are the unit the model must generalise over, and no augmentation manufactures a viewpoint.

WHAT THE STATISTIC RETURNED

- **1 or 2, in all six**
  three detectors × two threshold sets
- **Never 5**
  the value the PRD assumed

Fixing the statistic before seeing the data is what makes this a measurement rather than a preference. It was also chosen so that a wrong answer would favour keeping the PRD's value.

Two frozen backbones, arbitrated by a learned gate

MFSTNet is the project's core contribution. The gate is a research artifact in its own right — logged every run and analysed in the paper.

ResNet-50
2048 × 7 × 7
frozen

DINOv2 ViT-S/14
384 × 16 × 16
frozen

bidirectional
cross-attention
two directions

the gate
g = σ(W[Z_A;Z_B])
F = gZ_A + (1−g)Z_B

per-lane ROI
Voronoi cells from
lane centres

BiLSTM + temporal
attention, T = 60

congestion head
LOW / MED / HIGH
60 s ahead

Freezing the backbones and caching their features is what makes an eight-config ablation feasible on a 6 GB laptop GPU. Ablation-ability is treated as an architectural constraint, not an afterthought: every module can be disabled by a config flag, with no code edit between configs.

The gate can gate — the flat gate is a fact about the data

On the pilot corpus it sits at 0.54 with a spread of 0.005: one value everywhere, which is a scalar blend rather than a gate. So we asked whether it can do more.

SEED	GATE STD	G WHEN CNN CARRIES	G WHEN VIT CARRIES	SEPARATION
1	**0.348205**	0.41444	0.65744	0.243003
7	**0.324192**	0.44837	0.68359	0.235215
42	**0.451591**	0.44232	0.63055	0.188235

Given a task where half the samples carry the label in the CNN branch and half in the ViT, the gate moves by 50–90× its pilot spread and routes toward whichever branch is worth reading. The mechanism works; the pilot corpus simply does not demand it.

WHY IT IS FLAT, EXACTLY

- **z_a, z_b nearly orthogonal**
  `cosine −0.004 — different information`
- **but small: std 0.035**
  `so the sigmoid stays linear`
- **predicted gate std 0.006**
  `observed 0.006`

The residual adds both branches back ungated, so neither can be fully suppressed. That bounds how much of the ablation's gain gating can ever explain — which belongs in the paper rather than in a reviewer's question.

Baselines reported as measured, including where they disqualify

Thirty paired seeds per regime, bootstrap confidence intervals, paired t-test and Cohen's d — the statistics the requirements ask for.

REGIME	FIXED-TIME	LONGEST-QUEUE	WEBSTER	COHEN'S D	CITABLE
light	11.92	6.28	6.59	15.99	partly
saturated	**26.18**	**27.94**	**14.05**	**2.94**	yes
oversaturated	71.89	70.46	66.61	0.31	no

mean wait, seconds · headline comparison is made at saturation

TWO AUTOMATIC DISQUALIFICATIONS

- **Light: Webster clamps 100%**
  a fixed-time controller in disguise
- **Oversaturated: <85% arrive**
  a low wait measures survivors only

Both rules were written before the runs. They exist because the flattering reading of each regime is available and wrong — a controller that finishes fewer trips can post a better mean wait simply by losing the slow ones.

The ablation went from 12.3 hours to 1.5

Two independent faults were hiding behind one symptom, and half of every training epoch was spent on the wrong side of the bus.

CONFIG	CPU	GPU FP32	GPU FP16	GAIN
A · CNN only	33.2 s	20.5 s	12.7 s	2.6×
B · ViT only	20.1 s	19.8 s	9.4 s	2.1×
G · full MFSTNet	208.4 s	40.0 s	16.6 s	12.6×
A–H, extrapolated	**12.3 h**	**3.8 h**	**1.5 h**	**8.2×**

measured on the five-camera pilot during the optimisation work; not a committed CSV

- **The documented install could not work**
  It pointed at a CUDA index that no longer carries PyTorch at any version, so it silently resolved to the CPU wheel.

- **The trainer had no device handling at all**
  A correct install would still have run on the CPU with the GPU idle. It does not error; it is just slow on a machine that looks healthy.

- **Features were widened too early**
  The cache stores fp16 and the model wants fp32. Widening before the transfer doubled what was copied and shipped — and the widening is exact, so moving it changed no result.

REPRODUCIBILITY MACHINERY

- **Pre-registered thresholds**
  fixed before the run that tests them
- **Results written, never typed**
  a short run refuses to overwrite a long one
- **The corpus audits itself**
  9 checks; 7 of 7 mutations caught
- **Results carry their environment**
  torch build, device, TF32, commit

629 tests · 179 commits · 18 decision records · seeds fixed at 42 across Python, NumPy, PyTorch and SB3.

What went wrong, and what happens now

Twenty-nine defects are logged with their cause and cost. These are the ones that changed the project.

#	WHAT WAS WRONG	WHY IT MATTERED
P21	e_rickshaw scored 0.000	IDD contains zero instances — a head trained on nothing
P23	Lane polygons overlapped on 11 of 12 cameras	every vehicle in the overlap double-counted
P26	Every camera pooled through one camera's lanes	silent: shapes agree, loss falls, wrong region
P29	A cached window read its entire clip	26 GB of churn for 1.5 GB — segfault, no traceback
P30	No pinned package matched its installed version	a clean machine would build a different stack
P34	Calibration produced an impossible threshold	0% LOW — the degeneracy it exists to prevent

Three of these shared one shape — a number whose meaning depended on a fact the number did not carry. Each was fixed where it happened; the pattern was then closed once, by recording the environment beside every result.

THREE DECISIONS, THEN THE CORPUS

- **A28 — sampling interval**
  eleven cameras, or seven
- **ADR-017 — thresholds**
  what every congestion table means
- **S17 — acceptance surface**
  which detector is the one of record

Everything on the critical path is built and measured. What remains is not engineering: place the lane centres, build the corpus, audit it, run the ablation — an evening's compute.

Honest limit: 64 independent segments across 11 cameras, and 20 of them come from one clip. That concentration is the corpus's real weakness.

Sources, and what each one obliges us to do

Written down now rather than reconstructed in Week 20. Two of these carry conditions that shape what can be released.

IDD	Varma, G., Subramanian, A., Namboodiri, A., Chandraker, M., & Jawahar, C. V. (2019). IDD: A Dataset for Exploring Problems of Autonomous Navigation in Unconstrained Environments. WACV 2019.	• Citation required by the terms of use.
TrafficCAM	Deng, Z., Cheng, Y., Liu, L., Wang, S., Ke, R., Schönlieb, C.-B., & Aviles-Rivero, A. I. TrafficCAM: A Versatile Dataset for Traffic Flow Segmentation. arXiv:2211.09620. University of Cambridge & University of Bristol.	• No licence declared — cite, and seek written permission before release.
ITD v1.2	Agarwal, Thombre, Kedia & Ghosh (2024). Indian Traffic Dataset. Transportation Engineering Group, IIT Roorkee. CC BY-NC 4.0.	• Non-commercial: the restriction propagates to anything derived.
BMD-45	IISc AIM. BMD-45 (Hugging Face: iisc-aim/BMD-45). CVPR 2026. arXiv:2604.24419. CC BY 4.0.	• Author list to be completed from the paper before submission.
Bellevue Traffic Video Dataset	City of Bellevue, Washington. TrafficVideoDataset. github.com/City-of-Bellevue/TrafficVideoDataset.	• Released by a public body for research use.
Tooling	Jocher et al., Ultralytics YOLOv8 · Oquab et al. (2024), DINOv2 · Raffin et al. (2021), Stable-Baselines3, JMLR 22(268) · Lopez et al. (2018), Microscopic Traffic Simulation using SUMO, ITSC.	• Cited in the paper's implementation section.

Full entries in docs/00-planning/DATASETS.md · one marked item is deliberately incomplete rather than guessed